

3rd year

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ذكاء شالتي

Signals & Systems Lab

Lab (6)

Properties of Signals 2

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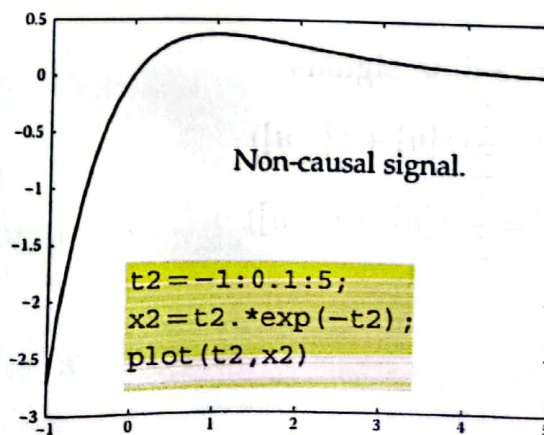
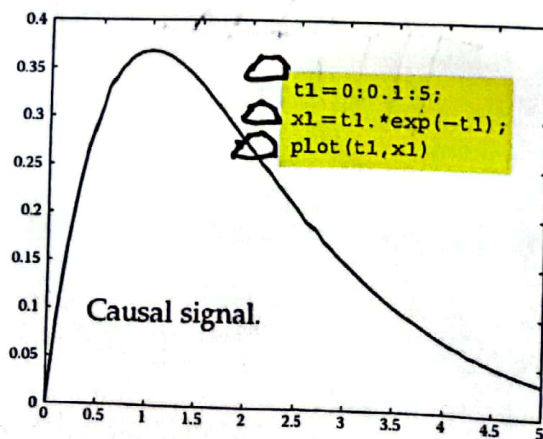
Causal Signals

Causal, non-causal, anti-causal

A signal $x(t)$ (or $x[n]$) is causal if it is zero for all negative values of time, i.e., $x(t) = 0, t < 0$ (or $x[n] = 0, n < 0$).

Non-causal signals are signals that have nonzero values in both positive and negative time.

Finally, a signal that is zero for all positive times is called anti-causal.



Even and Odd Signals



even (or has even symmetry or is an even function of t)

$$x(-t) = x(t), \quad -\infty < t < \infty. \quad x[n] = x[-n] \quad \text{discrete-time signals}$$

odd (or has odd symmetry or is an odd function of t)

$$x(-t) = -x(t), \quad -\infty < t < \infty. \quad -x[n] = x[-n] \quad \text{discrete-time signals}$$

Example

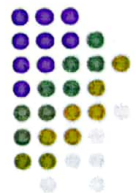
Find out if the signals $x(t) = t^2$ and $y(t) = t^3$ are even or odd.

symmetry of $x(t)$ and $y(t)$ is tested in the time interval $-4 \leq t \leq 4$.

Commands	Results						Comments
t1=0:4	t1 = 0	1.00	2.00	3.00	4.00		The time t .
t2=0:-1:-4	t2 = 0	-1.00	-2.00	-3.00	-4.00		The time $-t$.
x1=t1.^2	x1 = 0	1.00	4.00	9.00	16.00		The signal $x(t)$.
x2=t2.^2	x2 = 0	1.00	4.00	9.00	16.00		The signal $x(-t)$. It is clear that $x(t) = x(-t)$, hence, $x(t) = t^2$ is an even signal.
y1=t1.^3	y1 = 0	1.00	8.00	27.00	64.00		The signal $y(t)$.
y2=t2.^3	y2 = 0	-1.00	-8.00	-27.00	-64.00		The signal $y(-t)$. It is clear that $-y(t) = y(-t)$, hence, $y(t) = t^3$ is an odd signal.



Even and Odd Signals



any signal $x(t)$ is expressed as $x(t) = x_e(t) + x_o(t)$, where

$$x_e(t) = \frac{1}{2} [x(t) + x(-t)]$$

$$x_o(t) = \frac{1}{2} [x(t) - x(-t)].$$

Even

Odd

discrete-time signals

$$x_e[n] = \frac{1}{2} (x[n] + x[-n])$$

$$x_o[n] = \frac{1}{2} (x[n] - x[-n])$$



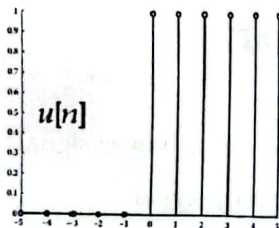
Even and Odd Signals

Example

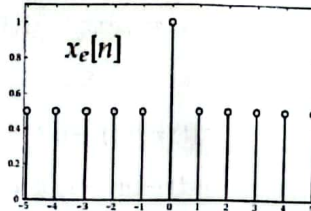
Separate the unit step sequence $u[n]$, $-5 \leq n \leq 5$ in even and odd parts. Verify your outcome by constructing the unit step sequence from its even and odd parts.

$$x[-n] = u[-n] = \begin{cases} 1, & n \leq 0 \\ 0, & n > 0 \end{cases}$$

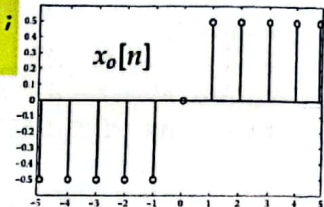
`n = -5:5;`
`u = (n >= 0);`
`stem(n, u);`



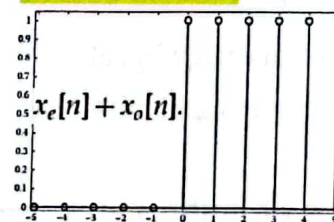
`u_n = (n <= 0)`
`ue = 1/2 * (u + u_n);`
`stem(n, ue);`



`uo = 1/2 * (u - u_n);`
`stem(n, uo);`



`stem(n, ue + uo)`



Energy and Power Signals

The energy E_x of a continuous-time signal $x(t)$

$$E_x = \lim_{T \rightarrow \infty} \int_{-T}^T |x(t)|^2 dt = \int_{-\infty}^{\infty} |x(t)|^2 dt, \quad |x(t)| \text{ absolute value of } x(t)$$

$$0 < E_x < \infty \Rightarrow x(t) \text{ is an energy signal}$$

The power P_x of a signal $x(t)$ is given by

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt.$$

If $x(t)$ is periodic, the power is computed from the relationship

$$P_x = \frac{1}{T} \int_{t_0}^{t_0+T} |x(t)|^2 dt.$$

$$0 < P_x < \infty \Rightarrow x(t) \text{ is called a power signal}$$

`limit(F, x, a)` computes the limit of the function F when the symbolic variable x tends to a .

Energy and Power Signals

Example

Verify that $x(t) = u(t)$ is a power-type signal while $x(t) = u(t) - u(t-1)$ is an energy-type signal.

$$x(t) = u(t)$$

```
syms t T
x=heaviside(t);
d=int(abs(x)^2,t,-T,T);
Ex=limit(d,T,inf)
Ex=Inf
Px=limit((1/(2*T))*d,T,inf)
Px=1/2
```

$u(t)$ is not an energy signal.

$u(t)$ is a power signal.

$$x(t) = u(t) - u(t-1)$$

```
syms t T
x=heaviside(t)-heaviside(t-1);
d=int(abs(x)^2,t,-T,T);
Px=limit((1/(2*T))*d,T,inf)
Px=0
Ex=limit(d,T,inf)
Ex=1
```

$x(t) = u(t) - u(t-1)$ is not a power signal.

$u(t) - u(t-1)$ is an energy signal.

signal is of finite energy $\Rightarrow$ the signal power is zero

signal is a power signal $\Rightarrow$ its energy is infinite



Energy and Power Signals

signal is of finite energy $\Rightarrow$ the signal power is zero

signal is a power signal $\Rightarrow$ its energy is infinite

$x(t) = t \Rightarrow$ neither energy signals nor power signals.

energy or the power of a signal in a definite time interval $t_1 \leq t \leq t_2$

$$E_x = \int_{t_1}^{t_2} |x(t)|^2 dt \quad P_x = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} |x(t)|^2 dt$$

Example

Compute the energy and the power of the signal $x(t) = 2\cos(\pi t)$ in one period time.

```
t1=0;
t2=2;
syms t
x=2*cos(pi*t);
Ex=int(abs(x)^2,t,t1,t2)
```

$$T = 2\pi/\Omega = 2\pi/\pi = 2.$$

$$E_x = 4$$

$$P_x = 2$$



Energy and Power Signals

The energy and power of a discrete-time signal

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2 = \lim_{N \rightarrow \infty} \sum_{n=-N}^N |x[n]|^2 \quad P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$$

Example

Compute the energy of the signal $x[n] = \cos(n)$, $-5 \leq n \leq 5$.

```
N=5;  
n=-N:N;  
x=cos(n);  
E=sum(abs(x).^2)      E=4.9058  
P=1/(2*N+1)*E         P=0.4460
```



Deterministic and Stochastic Signals

deterministic $\Rightarrow$ no randomness is involved in its values

each value is fixed and can be determined by a mathematical expression

example unit step function

stochastic or random $\Rightarrow$ randomness and its values cannot be predicted

example thermal noise produced by electronic devices

Thermal noise $\Rightarrow$ modeled as a sequence of random numbers

normal distribution

mean value $\mu = 0$

variance $\sigma^2 = 1$

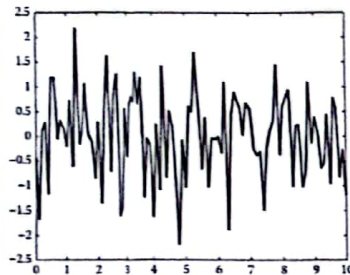


Deterministic and Stochastic Signals

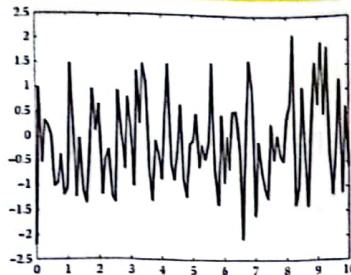
Thermal noise $\Rightarrow$ modeled as a sequence of random numbers

normal distribution
mean value $\mu = 0$
variance $\sigma^2 = 1$

```
t=0:.1:10;  
x1=randn(size(t));  
plot(t,x1);
```

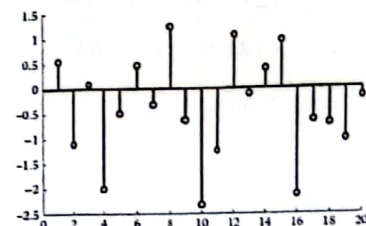


```
t=0:.1:10;  
x2=randn(size(t));  
plot(t,x2);
```



Graph of a stochastic signal $x_1(t)$. The signal is completely random as it does not depend on t , and its future values cannot be predicted.

```
x=randn(20,1);  
stem(x);
```



Random discrete-time signal.

